WORKSHEET

Chapter 1: Resultant and unit vectors in 3D space

1 If $\mathbf{a} = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - \mathbf{k}$ and $\mathbf{c} = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$, calculate:

a $|\mathbf{a}|$

b $|\mathbf{b}|$

c $|\mathbf{c}|$

d $\mathbf{a} + \mathbf{b}$

e $\mathbf{a} - 2\mathbf{b}$

f $2\mathbf{a} + \mathbf{b} - \mathbf{c}$

g $\mathbf{a} - \mathbf{b} + 3\mathbf{c}$

h $3\mathbf{a} + 4\mathbf{b} + \mathbf{c}$

2 Find the norm of each of the following vectors.

a $7\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$

b $7\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

c $3\mathbf{j} - 4\mathbf{k}$

d $2\mathbf{i} - \mathbf{j} + 5\mathbf{k}$

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e $2\mathbf{j} + 8\mathbf{k}$

f $-\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

3 Find the magnitude of each of the following vectors

a $2\mathbf{i} + 4\mathbf{j} + 3\mathbf{k}$

b $5\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

c $2\mathbf{j} - \mathbf{k}$

d $5\mathbf{i}$

e $3\mathbf{i} - 2\mathbf{j} - \mathbf{k}$

f $\mathbf{i} + \mathbf{j} + \mathbf{k}$

4 Points D, E and F have coordinates $(9, 1, 0)$, $(8, -3, 5)$ and $(5, 5, 7)$ respectively.

a Find the position vectors of D, E and F in $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ form.

b Find $\mathbf{DE}$ and $\mathbf{EF}$.

c Find $|\mathbf{DE}|$ and $|\mathbf{EF}|$.

5 Points A, B and C have coordinates $(-1, 1, 4)$, $(8, 2, 0)$ and $(5, -2, 11)$ respectively.

a Find the position vectors of A, B and C in $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ form.

b Find $\mathbf{AB}$ and $\mathbf{BC}$.

c Find $|\mathbf{AB}|$ and $|\mathbf{BC}|$.

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6 In each case below, calculate the unit vector in the direction of $\mathbf{a}$ ($\hat{\mathbf{a}}$).

a $\mathbf{a} = \mathbf{i} + 2\mathbf{j}$

b $\mathbf{a} = 2\mathbf{i} - 3\mathbf{k}$

c $\mathbf{a} = 3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$

d $\mathbf{a} = 4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$

e $\mathbf{a} = 5\mathbf{i} + \mathbf{j} - 7\mathbf{k}$

f $\mathbf{a} = 2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k}$

7 Determine the vector $\mathbf{PQ}$ in $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ form for each of the following pairs of points:

a $P\,(3, 7, 2)$ and $Q\,(9, 12, 5)$

b $P\,(4, 1, 0)$ and $Q\,(3, 4, -2)$

c $P\,(2, -3, 1)$ and $Q\,(0, -5, 3)$

d $P\,(9, 3, -2)$ and $Q\,(1, 3, 4)$

e $P\,(0, 1, 2)$ and $Q\,(-2, 1, 2)$

f $P\,(4, 3, 2)$ and $Q\,(10, 9, 8)$

8 For each of the vectors found in Question 7, determine a unit vector in the direction of $\mathbf{PQ}$

a

b

c

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d

e

f

9 For each of the following vectors, select any pairs that are parallel.

a $= \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

b $= 2\mathbf{i} - 4\mathbf{j} + 6\mathbf{k}$

c $= 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

d $= 4\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$

e $= -2\mathbf{i} + 4\mathbf{j} - 6\mathbf{k}$

f $= 6\mathbf{i} - 15\mathbf{j} + 9\mathbf{k}$

g $= \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

h $= -2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$

i $= 3\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}$

10 Which of the following vectors are parallel with $\mathbf{a} = 4\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$?

a $\mathbf{m} = 4\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$

b $\mathbf{d} = 12\mathbf{i} + 6\mathbf{j} - 18\mathbf{k}$

c $\mathbf{f} = 6\mathbf{i} - 2\mathbf{j} - 4\mathbf{k}$

d $\mathbf{q} = -2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$

e $\mathbf{r} = \mathbf{i} - \frac{1}{2}\mathbf{j} - 3\mathbf{k}$

f $\mathbf{u} = 12\mathbf{i} - 8\mathbf{j} - 15\mathbf{k}$

g $\mathbf{t} = -8\mathbf{i} - 4\mathbf{j} + 12\mathbf{k}$

h $\mathbf{c} = \mathbf{i} + \frac{1}{2}\mathbf{j} - \frac{3}{2}\mathbf{k}$

i $\mathbf{n} = 2\mathbf{i} - \mathbf{j} - 3\mathbf{k}$

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Answers

1 **a** $\sqrt{14}$

b $\sqrt{5}$

c $\sqrt{14}$

d $3\mathbf{i} + 3\mathbf{j} - 3\mathbf{k}$

e $-3\mathbf{i} + 3\mathbf{j}$

f $\mathbf{i} + 5\mathbf{j} - 7\mathbf{k}$

g $8\mathbf{i} + 6\mathbf{j} + 5\mathbf{k}$

h $14\mathbf{i} + 10\mathbf{j} - 8\mathbf{k}$

2 **a** $\sqrt{62}$

b $\sqrt{62}$

c 5

d $\sqrt{30}$

e $\sqrt{68}$

f $\sqrt{14}$

3 **a** $\sqrt{29}$

b $\sqrt{30}$

c $\sqrt{5}$

d 5

e $\sqrt{14}$

f $\sqrt{3}$

4 **a** $\mathbf{d} = 9\mathbf{i} + \mathbf{j}, \mathbf{e} = 8\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}, \mathbf{f} = 5\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$

b $\mathbf{DE} = -\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}, \mathbf{EF} = -3\mathbf{i} + 8\mathbf{j} + 2\mathbf{k}$

c $|\mathbf{DE}| = \sqrt{42}$, $|\mathbf{EF}| = \sqrt{77}$

5 **a** $\mathbf{a} = -\mathbf{i} + \mathbf{j} + 4\mathbf{k}, \mathbf{b} = 8\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}, \mathbf{c} = 5\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$

b $\mathbf{AB} = 9\mathbf{i} - \mathbf{j} - 2\mathbf{k}, \mathbf{BC} = -3\mathbf{i} - 2\mathbf{j} + 9\mathbf{k}$

c $|\mathbf{AB}| = \sqrt{86}$, $|\mathbf{BC}| = \sqrt{94}$

6 **a** $\frac{1}{\sqrt{5}}(\mathbf{i} + 2\mathbf{j})$

b $\frac{1}{\sqrt{13}}(2\mathbf{i} - 3\mathbf{k})$

c $\frac{1}{3\sqrt{3}}(3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k})$

d $\frac{1}{\sqrt{29}}(4\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$

e $\frac{1}{5\sqrt{3}}(5\mathbf{i} + \mathbf{j} - 7\mathbf{k})$

f $\frac{1}{7}(2\mathbf{i} - 6\mathbf{j} - 3\mathbf{k})$

7 **a** $6\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$

b $-\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$

c $-2\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$

d $-8\mathbf{i} + 6\mathbf{k}$

e $-2\mathbf{i}$

f $6\mathbf{i} + 6\mathbf{j} + 6\mathbf{k}$

8 **a** $\frac{1}{\sqrt{70}}(6\mathbf{i} + 5\mathbf{j} + 3\mathbf{k})$

b $\frac{1}{\sqrt{14}}(-\mathbf{i} + 3\mathbf{j} - 2\mathbf{k})$

c $\frac{1}{\sqrt{3}}(-\mathbf{i} - \mathbf{j} + \mathbf{k})$

d $\frac{1}{10}(-8\mathbf{i} + 6\mathbf{k})$

e $-\mathbf{i}$

f $\frac{1}{\sqrt{3}}(\mathbf{i} + \mathbf{j} + \mathbf{k})$

9 $\mathbf{a}||\mathbf{e}, \mathbf{b}||\mathbf{g}$ and $\mathbf{f}||\mathbf{h}$

10 Vectors $\mathbf{d}$, $\mathbf{q}$, $\mathbf{t}$ and $\mathbf{c}$

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